

EXAM 3

Closed Book (Refer to the provided equations sheet)

Time: 90 minutes

1. A long vertical cylinder with radius R is rotating with constant angular velocity Ω within a large body of fluid.

Postulates:

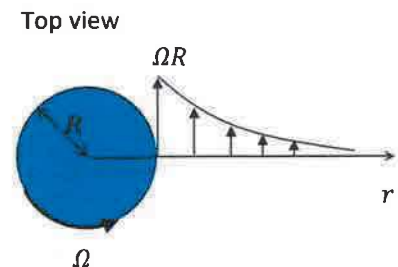
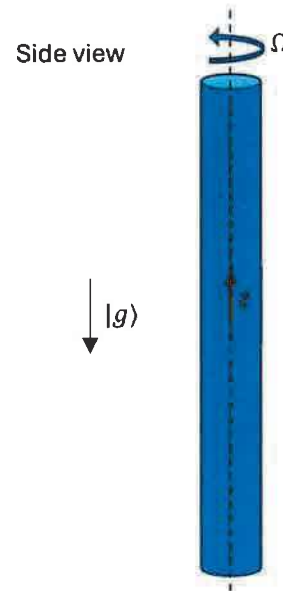
- Steady state, laminar flow.
- Newtonian fluid with constant viscosity μ and constant density ρ .
- Body of fluid around the cylinder is infinitely large.
- $v_r = 0$ and $v_z = 0$
- Pressure within the fluid is not a function of θ .

Answer the following questions in terms of parameters: ρ , μ , Ω , and R .

- 1.1. Using the mass balance (equation of continuity), demonstrate that v_θ is not a function of θ .
- 1.2. Using the equation of motion, derive an ordinary differential equation (ODE) for v_θ . Integrate the ODE to get an equation for v_θ as a function of r . Write down two boundary conditions for v_θ and evaluate the integration constants.

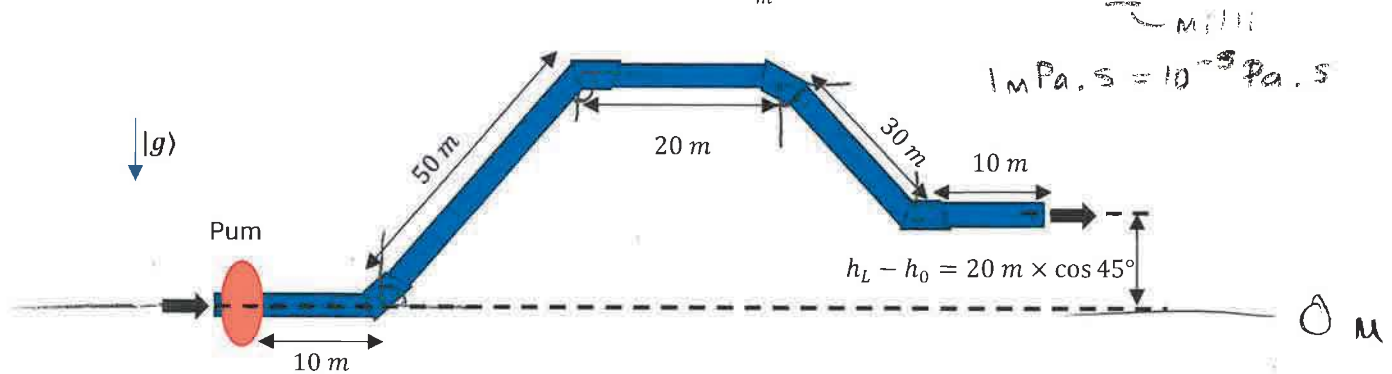
- 1) Integrate ODE
 2) Apply BC 1 $\rightarrow C_1$
 3) Integrate again v_θ
 4) Apply BC 2 $\rightarrow C_2$

$$v_\theta(r)?$$



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2. Water is pumped between two reservoirs at the rate of $10 \frac{\text{kg}}{\text{s}}$. The pipeline consists of a pump, five smooth circular pipes of 20 cm inner radius (lengths of which are provided in the schematic below), and four 45° elbows. The resistance of each elbow is roughly equivalent to that offered by a pipe whose length is 15 diameters. The density of water is $\rho = 1000 \frac{\text{kg}}{\text{m}^3}$ and its viscosity is $\mu = 1.06 \text{ mPa} \cdot \text{s}$.



- 2.1. Using the provided chart, determine the Fanning friction factor for the above system.
- 2.2. If both reservoirs are at atmospheric pressure ($1.01 \times 10^5 \text{ Pa}$), what is the pressure at the pump outlet?

starts on
page 7

$$Re = \frac{\rho \langle v \rangle D}{\mu}$$

$$F_k \equiv (2\pi RL) \left(\frac{1}{2} \rho \langle v \rangle^2 \right) f = (P_0 - P_L) \pi R^2$$

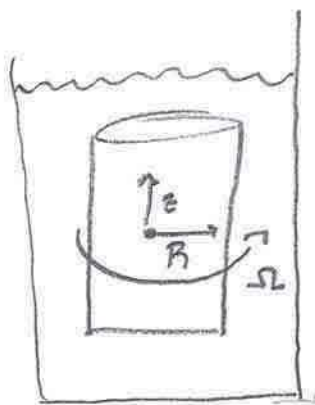
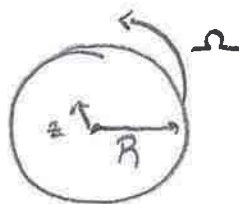
$$D \neq R$$

mass flow rate $\neq$ velocity

$$P_{in} \neq P_{out}$$

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- 1) Assumptions
 Steady state, $\frac{\partial}{\partial t} = 0$
 constant μ, ρ
 $R \rightarrow \infty$?



$v_r = 0 = v_z$
 P not a function of θ

$v_\theta(z, R, z)?$
 $v_\theta(r)$

1a) continuity equation (cylindrical)

$$\frac{\partial}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (r v_\theta) + \frac{\partial}{\partial z} (v_z) = 0$$

s.s. $v_r = 0$ $v_z = 0$
 $\frac{1}{r} \frac{\partial}{\partial \theta} (r v_\theta) = 0$ Axisymmetric? ✓

$\frac{\partial v_\theta}{\partial \theta} = 0$

B.C.s $\rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z} + \frac{v_r v_\theta}{r} \right) = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r} \frac{\partial^2 v_\theta}{\partial r \partial z} \right] + \rho g_\theta$

$$0 = \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} \right] + \rho g_\theta$$

$$0 = \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{\partial^2 v_\theta}{\partial z^2} \right] + \rho g_\theta$$

Not a function of z

$$0 = \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) \right]$$

$$0 = \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right)$$

$$\frac{\partial^2}{\partial r^2} (r v_\theta)$$

$\therefore v_\theta = C_1 r + \frac{C_2}{r}$ ✓

B.C. 1 no slip
 $v_\theta(r, r=R, z) = 0$ ✗
 $v_\theta(r, r=z) = 0$ ✗

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if $Re < 2100 \Rightarrow f = \frac{64}{Re}$?

2) $\dot{m} = 10 \frac{kg}{s}$

$Re = \frac{\rho \langle v \rangle D}{\mu}$

$r = 20 cm$

$D = 40 cm$

4 x 45° elbows

$\mu = 1.06 mPa \cdot s$

$P = 1000 \frac{kg}{m^3}$

$P = 1.01 \times 10^5 Pa$

$D = 2r = 2(20 cm) = 40 cm = 0.4 m$

$$\frac{1,000 \frac{kg}{m^3} (10 \frac{kg}{s}) (0.4 m)}{1.06 mPa \cdot s \times \frac{10^{-3} Pa \cdot s}{1 mPa \cdot s} - Pa \cdot s}$$

$$\frac{4,000 \frac{kg^2}{m^3 \cdot s}}{0.0106 \frac{kg}{m \cdot s^2}} \cdot \frac{1}{18 kg} = 20964.4$$

weight of water

$$\frac{P_o - P_L}{L}$$

$Re > 2100$ not laminar

$f = \frac{0.0791}{20964.4^{0.25}} = 0.574 \times 10^{-3}$

$g = 9.8 \frac{m}{kg}$

fin L of the line

$L = 10 + 50 \sin(45^\circ) + 20 + 30 \cos(45^\circ) + 10 = 96.56 m$

$h_L - h_o = 20 m \cos(45^\circ)$

if $h_o = 0$

$h_L = 20 m \cos(45^\circ) = 14.1421 m$

$\Delta P = f \left(\frac{L}{D} \right) \left(\frac{\rho \langle v \rangle^2}{2} \right) ?$

$$\frac{P_o - P_L}{L} = f \left(\frac{L}{D} \right) \left(\frac{\rho \langle v \rangle^2}{2} \right)$$

valid?
Where is gravity?
gravity cannot be ignored.

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